

Effects of Neonicotinoids Exposure on the Blue Crab (*Callinectes sapidus*): Bioaccumulation, Neurotoxicity, and Behavioral Implications

Pilar Gómez-Ramírez¹, Beatriz Fernández², Francisco Acosta-Cifuentes², Tatiana Oporto², Ignacio Franco², Livia Spadetto¹, Antonio Zamora-López³, Antonio Andrés Herrero-Reyes³, Pedro Javier Jiménez-Montalbán¹, Concepción Martínez-Gómez²

1 Area of Toxicology, Faculty of Veterinary, Campus de Espinardo, University of Murcia; 2 Oceanographic Centre of Murcia, Instituto Español de Oceanografía (IEO-CSIC); 3 Department of Zoology and Physical Anthropology, Faculty of Biology, University of Murcia

INTRODUCTION

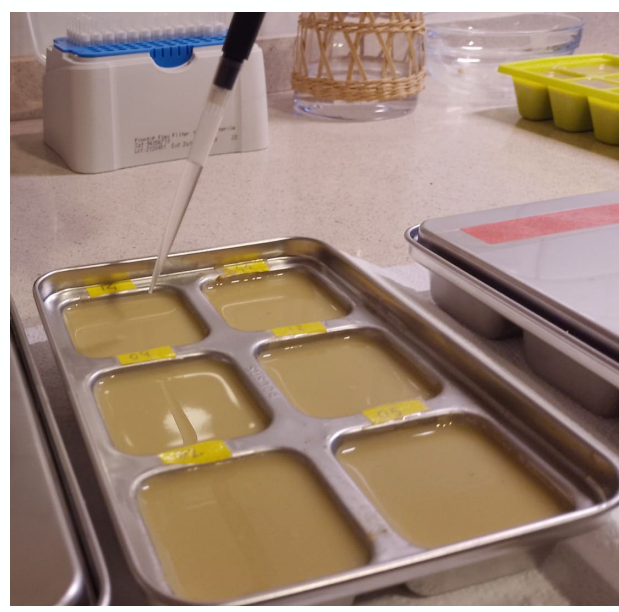
Crustaceans share more physiological characteristics with terrestrial arthropods than with fish or bivalves. There is limited information regarding the effects of neonicotinoid contamination on marine coastal areas affecting decapod crustaceans. This short-term experimental study investigates the neurotoxic effects after ingestion of food contaminated with the neonicotinoid pesticides ACETAMIPRID and IMIDACLOPRID, which are widely used in agriculture, in blue crabs (*Callinectes sapidus*).



MATERIAL AND METHODS



Collection and acclimation for 1 month



Oral exposure by seafood-like gelatin pellets spiked with the insecticides

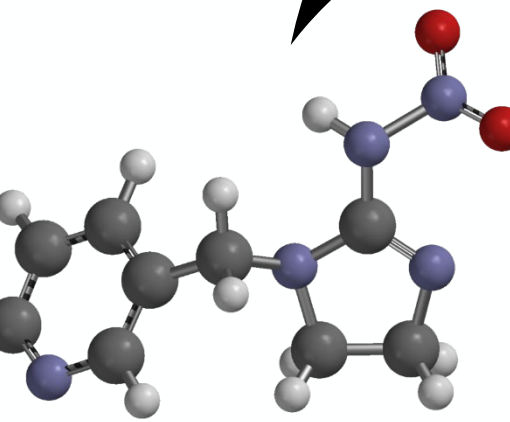
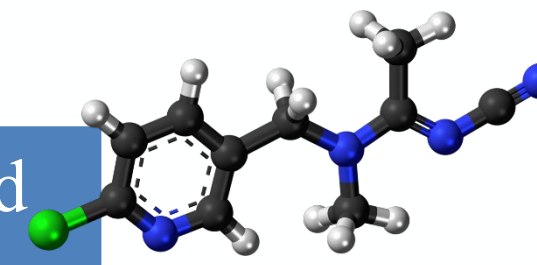
11♂+8♀ Control

6♂+6♀

Acetamiprid
3 µg/ g ww
14 d

6♂+6♀

Imidacloprid
1 µg/ g ww
23 d



- % Food intake
- Righting time
- AChE activity
- Other neurotoxicity signs

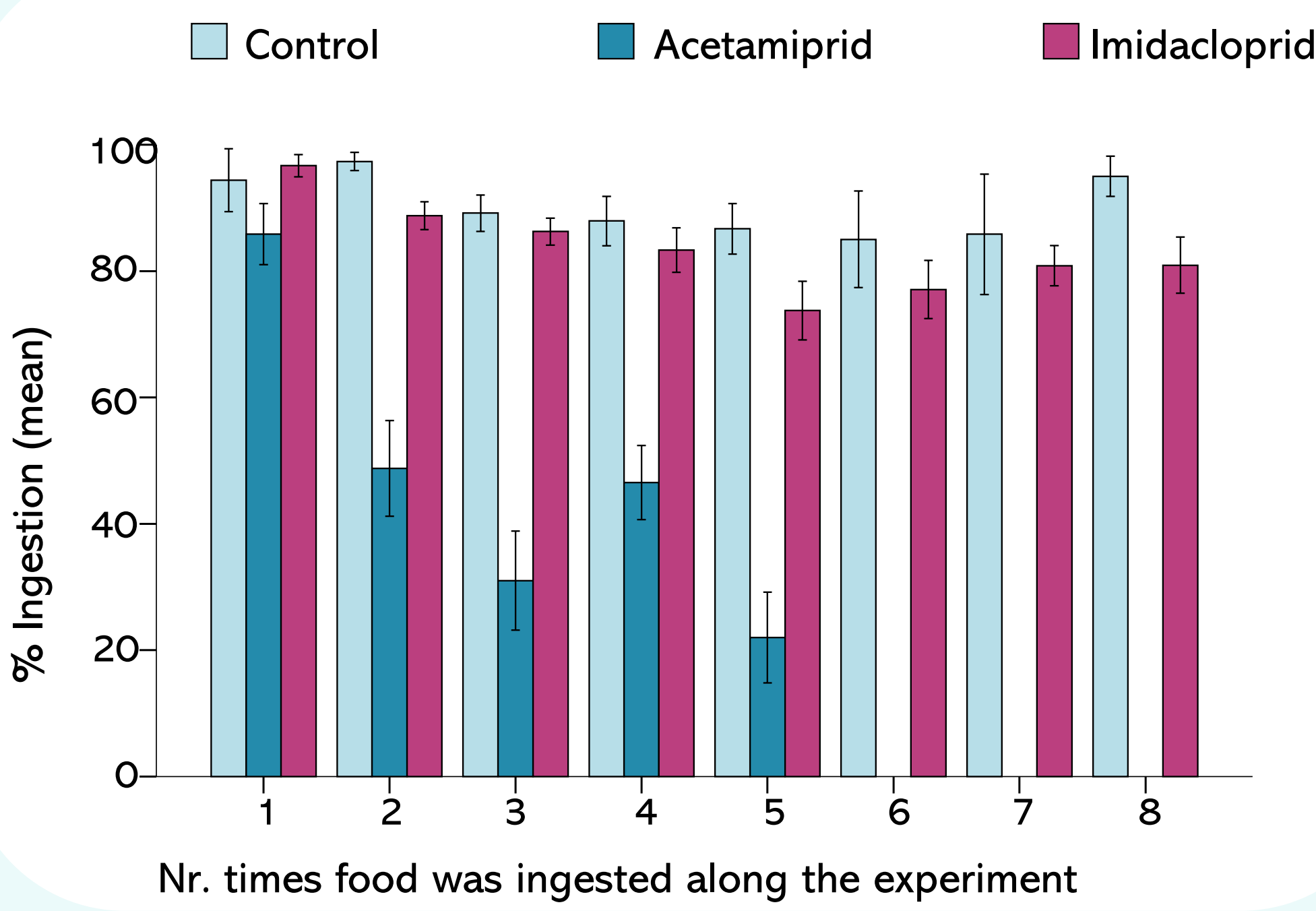


SPSS
AN IBM COMPANY

Pesticide tissue distribution

RESULTS AND DISCUSION

% Food intake



ACETAMIPRID-group showed a dramatic reduction of food intake after the first dose (from 86 to 25%). In some cases, food consumption ceased up to 11 days, before resuming to feed.

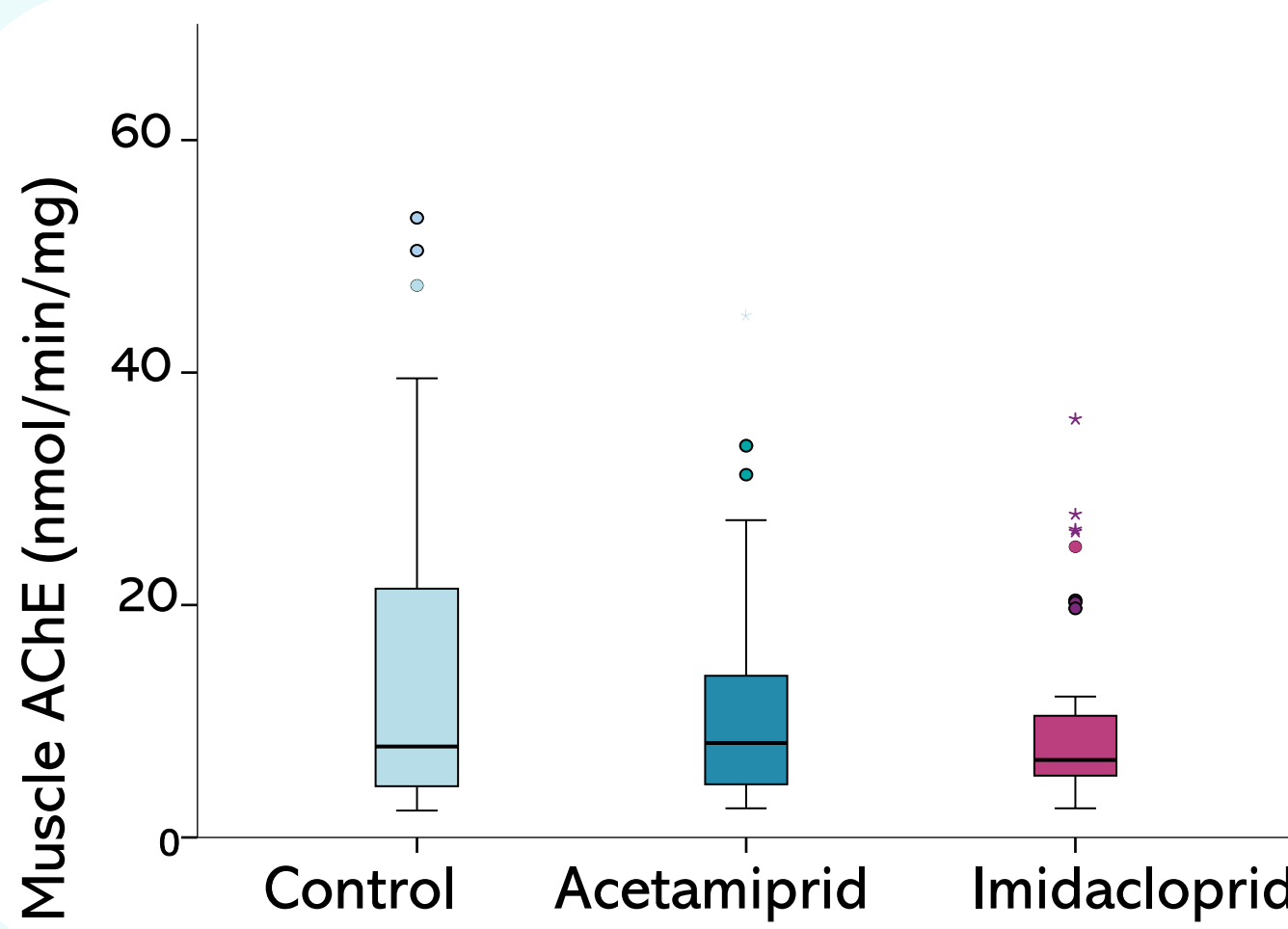
Pesticide tissue distribution

	ACETAMIPRID Median; mean±SE (range)	IMIDACLOPRID Median; mean±SE (range)
HAEMOLYMPH (ng mL ⁻¹)	4.48; 4.91±0.71 (1.76-10.50)	3.02; 2.99±0.59 (nd-6.38)
MUSCLE (ng g ⁻¹ ww)	6.61; 7.69±1.33 (0.47-15.07)	9.38; 12.74±2.42 (0.35-26.04)
HEPATOPANCREAS (ng g ⁻¹ ww)	17.87; 20.55 ±3.00 (9.28-45.12)	22.83; 22.41 ±1.94 (9.20-33.64)

Both insecticides were detected and quantified in all the samples, except IMIDACLOPRID in one haemolymph sample. ACETAMIPRID and IMIDACLOPRID followed the same pattern of tissue accumulation: HEPATOPANCREAS > MUSCLE > HAEMOLYMPH

AChE activity

AChE activity in muscle was lowest in the IMIDACLOPRID group. However, no statistically significant differences were found between the groups. Similar results have been found in other crab species exposed to neonicotinoids such as thiametoxam.

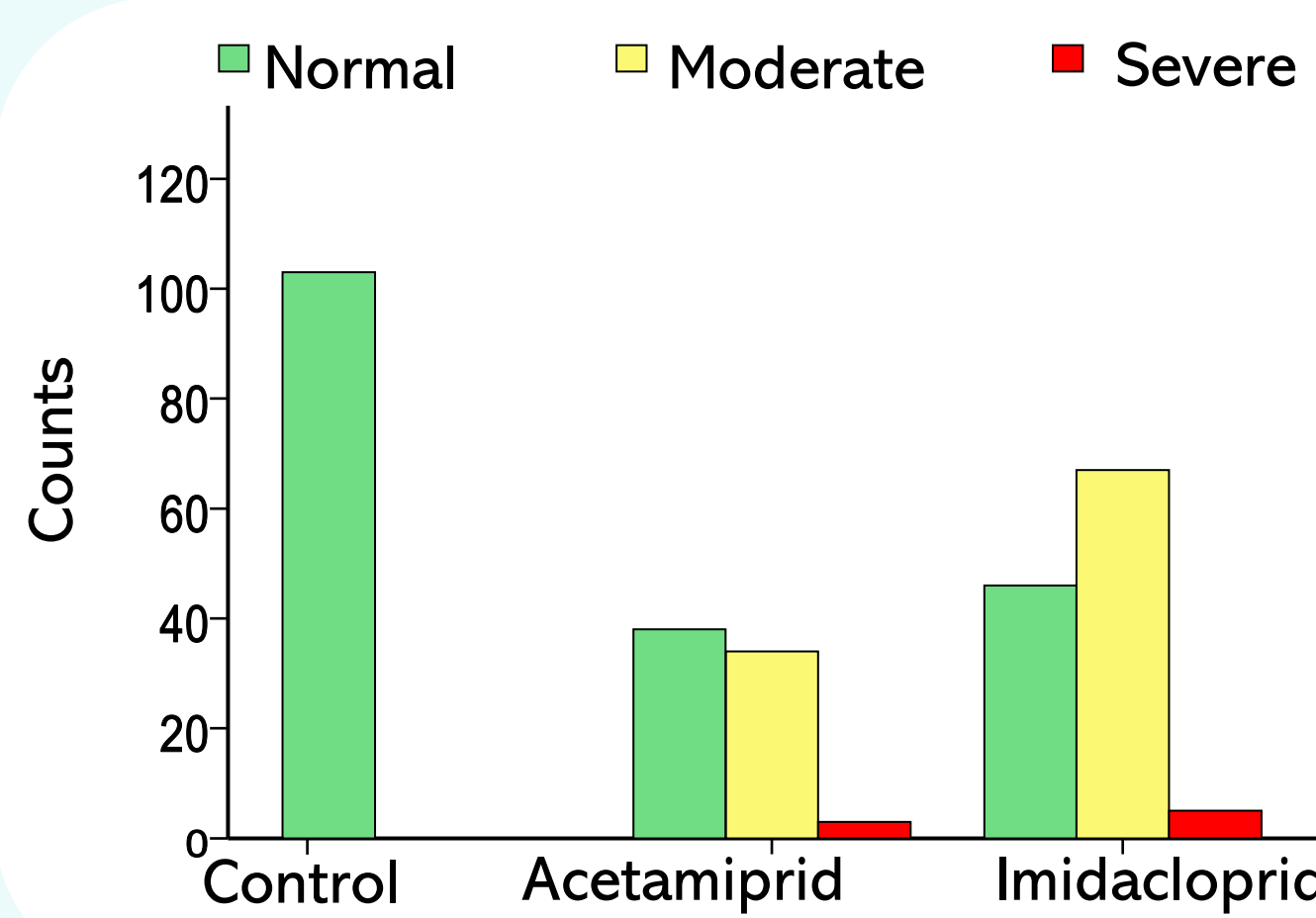


Righting time

The time a crab takes to flip-over when it is placed on the back was categorised as follows:

- ✓ Normal: ≤ 2 s
- ✓ Moderate effect: 2- 16 s
- ✓ Severe: > 16 s

A significant association between treatment and delayed righting time was found, finding more affection in IMIDACLOPRID-group



Other neurotoxicity signs

All individuals from ACETAMIPRID-group showed neurological effects after exposure:

- ✓ cheliped retraction disfunction
- ✓ chelae spasms
- ✓ sustained spastic leg movements



CONCLUSIONS

- ✓ Prolonged intake of IMIDACLOPRID and ACETAMIPRID caused bioaccumulation in HEPATOPANCREAS>MUSCLE>HAEMOLYMPH, and decreased AChE activity in muscle.
- ✓ The exposure to ACETAMIPRID and IMIDACLOPRID at the tested concentrations decreased crabs' vitality and caused a negative effect on predation capacity and defence ability.
- ✓ The use of blue crab as target species to investigate the effects of neonicotinoids in coastal marine ecosystems is promising.